

Practice questions Lecture 1 to 5

Question 1

Write the following expression in the form of $a + ib$.

$$\frac{3+i}{2-i}, \frac{4+10i}{-9+7i}$$

Solution

$$\text{a) } \frac{3+i}{2-i}$$

In order to simplify the above fraction, we will multiply and divide it by the conjugate of its denominator

$$\begin{aligned} \frac{3+i}{2-i} &\times \frac{2+i}{2+i} \\ &= \frac{6+3i+2i+(-1)}{4-(-1)} \\ &= \frac{5+5i}{5} = 1+i \end{aligned}$$

b) Part two will be done in the similar way. The simplified form will be

$$\frac{-1}{65}(-17+59i)$$

Question 2

$$z = -5+8i \text{ then find } z - z^*$$

Solution

$$z = -5+8i$$

$$z^* = -5-8i$$

$$\begin{aligned} z - z^* &= -5+8i - (-5-8i) \\ &= -5+8i + 5+8i = 16i \end{aligned}$$

Question 3

$$\text{If } z = 3+6i \text{ then find } z \cdot z^*$$

Solution

$$z = 3 + 6i$$

$$z^* = 3 - 6i$$

$$\begin{aligned} z \cdot z^* &= (3 + 6i)(3 - 6i) \\ &= 9 - 18i + 18i + 36 = 45 \end{aligned}$$

Question 4

Evaluate

$$\left| \frac{4 + 7i}{-5 + 6i} \right|$$

Solution

$$\begin{aligned} \frac{4 + 7i}{-5 + 6i} \cdot \frac{-5 - 6i}{-5 - 6i} &= \frac{(4 + 7i)(-5 - 6i)}{(-5 + 6i)(-5 - 6i)} \\ &= \frac{-20 - 24i - 35i - 42i^2}{(-5)^2 - (6i)^2} \\ &= \frac{22 - 59i}{25 + 36} = \frac{22 - 59i}{61} = \frac{22}{61} - \frac{59}{61}i \\ \left| \frac{4 + 7i}{-5 + 6i} \right| &= \sqrt{\left(\frac{22}{61}\right)^2 + \left(\frac{59}{61}\right)^2} = \sqrt{\frac{484 + 3481}{3721}} = \sqrt{\frac{3965}{3721}} \end{aligned}$$

Question 5

Write in set builder notation

1. A = Set of first 100 odd numbers
2. A = Set of Natural numbers starting from 2 to 40.
3. A = Set of all even numbers greater than 50.

Solution

$$A = \{x = 2k + 1 \mid 0 \leq k \leq 99\}$$

$$A = \{x : x \in \mathbb{N} \mid 2 \leq x \leq 40\}$$

$$A = \{x : x \in \mathbb{E} \wedge x > 50\}$$

Question 6

List the elements of the set

1. $A = \{x \mid x \in \mathbb{E}^+ \wedge x \leq 10\}$

2. $A = \{x \mid x^2 = 9, x \text{ is even}\}$
3. $A = \{x \mid x \in N \wedge 2 \leq x \leq 10\}$

Solution

$$A = \{2, 4, 6, 8, 10\}$$

$$A = \{\} = \phi = \text{Empty set}$$

$$A = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Question 7

Let A be the set of all even numbers.

Let B be the set of all Number less than 100.

Is A a subset of B? or

Is B a subset of A?

Solution

Neither A is a subset of B nor B is a subset of A.

Question 8

The complex numbers Z is given by

$$Z = \frac{i-1}{i}$$

Find modulus of Z .

Solution

$$Z = \frac{i-1}{i}$$

In order to find its modulus, we will first simplify it in the form of $a+ib$, for that we know that we are to multiply and divide it by the conjugate of denominator

$$= \frac{i-1}{i} \times \frac{-i}{-i} = \frac{i^2 - i}{i^2} = 1 + i$$

Now, we can find its absolute value or modulus as given below.

$$|Z| = |1+i| = \sqrt{1+1} = \sqrt{2}$$

Question 9

Evaluate

1. $(2+i)(3+i)$
2. $(4+5i)(-2-2i)$

3. $(-1-i)(-2-2i)$

4. i^3

Solution

$$(2+i)(3+i)$$

1) $= 6 + 2i + 3i - 1$

$$= 5 + 5i$$

2) It will be simplified in the similar pattern as above, the answer will be $-2(-1+9i)$

3) $4i$

4) $-i$

The Venn diagram given below shows the sets A, B, and C. Shade the following sets

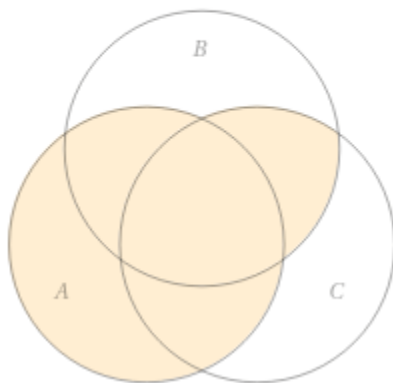
i. $A \cup (B \cap C)$

ii. $A \cup (B \cap C)^c$

iii. $A \cap (B \cup C)$

$$A \cup B \cap C$$

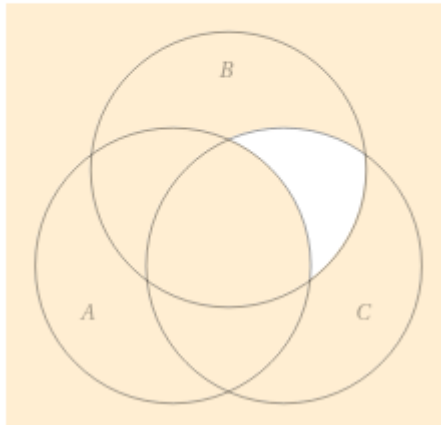
Venn diagram



Input interpretation

$$A \cup (B \cap C)'$$

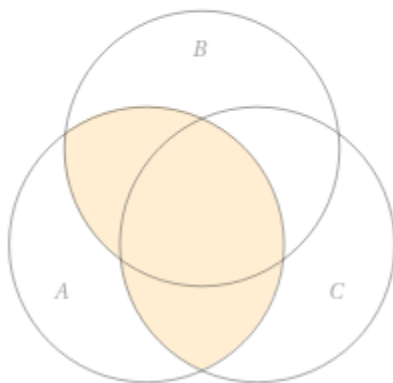
Venn diagram



Input interpretation

$$A \cap (B \cup C)$$

Venn diagram



Question 10

What is the cardinality of these sets?

A = Set of even integers

B = Set of prime numbers less than 30

Solution

Cardinality of A is equal to the cardinality of set of Natural Numbers
Cardinality of B is equal to 11

Question 11

- i) $z = -7 + 5i$
- ii) $z = 3i$

Find $z + z^*$, $z.z^*$, $z - z^*$

Solution

$$\begin{aligned} z + z^* &= -14 \\ \text{i)} \quad z.z^* &= 74 \\ z - z^* &= 10i \\ z + z^* &= 0 \\ \text{ii)} \quad z.z^* &= 9 \\ z - z^* &= 6i \end{aligned}$$

Question 12

Evaluate

$$\begin{aligned} \text{i.} \quad & \left| \frac{2+2i}{-5+3i} \right| \\ \text{ii.} \quad & \left| \frac{1+i}{2+3i} \right| \\ \text{iii.} \quad & \left| \frac{7+3i}{3+5i} \right| \end{aligned}$$

Solution

As is taught in the previous question, before finding the modulus, we will first simplify the expression in to the form of $a + ib$ and it will be done by multiply and dividing the given fraction by the conjugate of denominator.

$$\begin{aligned} \text{i)} \quad & \left| \frac{2+2i}{-5+3i} \right| = \left| \frac{-2}{17}(1+4i) \right| \\ & = \frac{2}{\sqrt{17}} \\ \text{ii)} \quad & \left| \frac{7+3i}{3+5i} \right| = \left| \frac{1}{17}(18-13i) \right| \\ & = \frac{\sqrt{493}}{17} \end{aligned}$$

Question 13

Write the domain and range of

$$\begin{aligned} \text{i.} \quad & f(x) = 3x \\ \text{ii.} \quad & f(x) = \frac{4}{3+2x} \\ \text{iii.} \quad & f(x) = \frac{2}{3-x} \\ \text{iv.} \quad & f(x) = \sqrt{5-2x} \end{aligned}$$

Solution

i) Domain = $\mathbb{R}$ = all real numbers
Range = $\mathbb{R}$ = all real numbers

ii) Domain = $\{x \in \mathbb{R} : x \neq \frac{-3}{2}\}$

Range = $\{y \in \mathbb{R} : y \neq 0\}$

iii) Domain = $\{x \in \mathbb{R} : x \neq 3\}$

Range = $\{y \in \mathbb{R} : y \neq 0\}$

iv) Domain = $\{x \in \mathbb{R} : x \leq \frac{5}{2}\}$

Range = $\{y \in \mathbb{R} : y \geq 0\}$

Question 14

Write the domain and range of $f(x) = 6x + 9$

Solution

Domain of $f(x)$ = All real no's

Range of $f(x)$ = All real no's

Question 15

Write the domain and range of $g(x) = \frac{5}{x+6}$

Solution

Domain of $g(x)$ = $\mathbb{R} - \{-6\}$

Range of $f(x)$ = All real numbers except zero.

Question 17

Determine whether the function is even, odd, or none.

i. $f(x) = x^2 + 3x$

ii. $f(x) = \sqrt{x^2 + 9}$

iii. $f(x) = x^2 + 4$

iv. $f(x) = \frac{x}{x^2 - 1}$

v. $f(x) = 6x^5 - x^3$

vi. $f(x) = 2x^4$

Solution

We know that a function is even if $f(x) = f(-x)$ and odd if $f(-x) = -f(x)$, so

i) $f(x) = x^2 + 3x$ Neither even nor odd

ii) $f(x) = \sqrt{x^2 + 9}$ Even function

iii) $f(x) = x^2 + 4$ Even function

iv) $f(x) = \frac{x}{x^2 - 1}$ Odd function

v) $f(x) = 6x^5 - x^3$ Odd function

vi) $f(x) = 2x^4$ Even function

Question 18

Prove that $f(x) = x^4 + 5x^2 + 3$ is an even function

Solution

$$f(x) = x^4 + 5x^2 + 3$$

$$f(-x) = (-x)^4 + 5(-x)^2 + 3$$

$$f(-x) = x^4 + 5x^2 + 3 = f(x)$$

Question 19

Prove that $f(x) = x^3 - 8x$ is an odd function

Solution

$$f(x) = x^3 - 8x$$

$$f(-x) = (-x)^3 - 8(-x)$$

$$f(-x) = -x^3 + 8x = -(x^3 - 8x) = -f(x)$$

Question 20

Express the quadratic polynomial in complete square form.

1. $2x^2 + 8x + 6$

2. $x^2 + 2x - 7$

3. $2x^2 + 6x + 3$

4. $x^2 - 2x + 1$

Solution

$$\begin{aligned}
 & 2x^2 + 8x + 6 \\
 1) &= 2(x^2 + 4x + 3) \\
 &= 2(x^2 + 2(x)(2) + 4 - 4 + 3) \\
 &= 2[(x+2)^2 - 1] \\
 2) & (x+1)^2 - 8 \\
 3) & 2\left(x + \frac{3}{2}\right)^2 - \frac{3}{2} \\
 4) & (x-1)^2
 \end{aligned}$$

Question 21

Express the quadratic polynomial $x^2 + 4x + 6$ in complete square form.

Solution

$$x^2 + 4x + 6 = x^2 + 4x + 4 - 4 + 6 = (x+2)^2 + 2$$

Question 22

1. If the functions f and g are defined by

$$f(x) = x^2 \text{ and } g(x) = 2x + 3$$

Find $g(f(x))$.

2. If the functions f and g are defined by

$$f(x) = 2x + 5 \text{ and } g(x) = x^2 + 5$$

Find $f(x) + g(x)$, $f(x) - g(x)$, $(f - g)(1)$, $f(g(x))$, $(f \cdot g)(x)$

Solution

$$\begin{aligned}
 1. & 4x^2 + 12x + 9 \\
 2. & f + g = x^2 + 2x + 10 \\
 & f - g = -x^2 + 2x \\
 & (f - g)(1) = 1 \\
 & f[g(x)] = 2x^2 + 15 \\
 & f \cdot g = 2x^3 + 5x^2 + 10x + 25
 \end{aligned}$$

Question 23

If the functions f is defined by

$$f(x) = \frac{1}{x}, \quad 0 < x \leq 5,$$

then find the range of f .

Solution

Since $x \leq 5$, $1/x \geq 1/5$

i.e.,

$$f(x) \geq 1/5.$$

Question 24

If the functions f and g are defined by

$$f(x) = \frac{3}{x}, \quad 0 < x \leq 3,$$

$$g(x) = 5x - 1, \quad x \in R,$$

then Calculate $gf(2)$.

Solution

$$gf(x) = g(f(x)) = g(3/x) = 5(3/x) - 1 = (15/x) - 1.$$

So,

$$gf(2) = (15/2) - 1 = 13/2$$

Question 25

If the functions g is defined by

$$g(x) = 8x - 5, \quad x \in R,$$

then find an expression in terms of x for $g^{-1}(x)$.

Solution

To find $g^{-1}(x)$, we write $y = 8x - 5$

$$y + 5 = 8x$$

$$x = (y + 5)/8.$$

Therefore,

$$g^{-1}(x) = (x + 5)/8$$

Question 26

Let $f(x) = 6x + 1$ and $g(x) = -9x + 5$. Find the product of f & g i.e $(f.g)(x)$.

Solution

$$\begin{aligned}(f.g)(x) &= f(x).g(x) = (6x+1).(-9x+5) \\ &= -54x^2 + 21x + 5 \quad (\text{Ans.})\end{aligned}$$

Practice Questions for Lecture 6 to 10

Question 1

Determine whether the function is even, odd, neither even nor odd.

- $f(x) = x^2 + 3x$
- $f(x) = \sqrt{x^2 + 9}$
- $f(x) = x^2 + 4$
- $f(x) = \frac{x}{x^2 - 1}$
- $f(x) = 6x^5 - x^3$
- $g(x) = x^4 + x^6$

Question 2

Express the quadratic polynomial in complete square form

$$2x^2 + 8x + 6$$

Question 3

If $f(x) = 4x - 3$ & $g(x) = x^3 + 3$

Then find $g \circ f(x)$ and $f \circ g(x)$.

Question 4

Let f and g be the functions defined by

$$f(x) = 3x + 3$$

$$g(x) = 3x + 5$$

then find $f \circ g(x)$ and $g \circ f(x)$.

Solution

$$f \circ g(x) = f(g(x))$$

$$= f(3x + 5)$$

$$= 3(3x + 5) + 3$$

$$= 9x + 18$$

$$g \circ f(x) = g(f(x))$$

$$= g(3x + 3)$$

$$= 3(3x + 3) + 5$$

$$= 9x + 14$$

Question 5

Determine whether the following equations have one, two or no real solution.

- $7x^2 + 9x - 3 = 0$
- $4x^2 + 2x + 1 = 0$

Question 6

Determine whether the equation has one, two or zero real solution.

$$5x^2 + 4x - 6 = 0$$

solution:

$$a = 5, b = 4, c = -6$$

$$b^2 - 4ac = 16 + 120 = 136 > 0$$

So there will be two distinct real solutions.

Question 7

Determine whether the equation has one, two or zero real solution.

$$x^2 + 3x + 6 = 0$$

solution :

$$a = 1, b = 3, c = 6$$

$$b^2 - 4ac = 9 - 24 = -15 < 0$$

So there will be no real solutions.

Question 8

If the following equation has one solution then find the value of a

$$ax^2 + 5x + 2 = 0$$

Question 9

The equation $tx^2 + 5x - 6 = 0$ has two real solutions. What can be deduced about the value of t?

Solution

$$tx^2 + 5x - 6 = 0$$

$$a = t, b = 5, c = -6$$

For two real solutions the discriminant should be > 0 i.e

$$b^2 - 4ac > 0$$

$$25 + 24t > 0$$

$$24t > -25$$

$$t > \frac{-25}{24}$$

Question 10

The equation $kx^2 - 8x + 9 = 0$ has exactly one solution. What can be deduced about the value of k?

Solution

$$kx^2 - 8x + 9 = 0$$

$$a = k, b = -8, c = 9$$

For exactly one solution the discriminant should be equal to 0 i.e

$$b^2 - 4ac = 0$$

$$64 - 36k = 0$$

$$64 = 36k$$

$$\frac{64}{36} = k$$

$$\frac{32}{18} = k$$

Question 11

Given matrices A and B such that

$$A = \begin{bmatrix} 1 & 2 \\ -3 & -6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ -2 & -4 \end{bmatrix}$$

Find $2A - 3B$.

Question 12

Check whether the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ can be multiplied? Justify your answer.

Question 13

Find the inverse $A = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix}$

Question 14

Multiply the matrix:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Question 15

Find the transpose of matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -3 & -2 & 9 \\ -5 & 7 & -3 \end{bmatrix}$$

Practice Questions for Lecture 11 to 15

Question 1

Define Echelon form and reduced echelon form.

Question 2

If $A = \begin{bmatrix} 2 & -2 & 1 \\ 2 & 6 & 5 \\ 0 & 0 & 1 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then find $A^2 - 3I$.

Where I is the identity matrix of order 3×3 .

Answer

$$\begin{bmatrix} -3 & -16 & -7 \\ 16 & 29 & 37 \\ 0 & 0 & -2 \end{bmatrix}$$

Question 3

Find the matrix of cofactors of the given matrix

$$C_{ij} = \begin{bmatrix} -2 & -1 & 1 \\ 5 & -3 & 5 \\ 4 & 6 & 2 \end{bmatrix}$$

Answer

$$C_{ij} = \begin{bmatrix} -36 & 10 & 42 \\ 8 & -8 & 8 \\ -2 & 5 & 11 \end{bmatrix}$$

Question 4

For the matrices A and B such that

$$A = \begin{bmatrix} -4 & -2 \\ 9 & 8 \end{bmatrix} \quad B = \begin{bmatrix} 0 & -5 \\ 5 & 8 \end{bmatrix}$$

Evaluate $A - 3B$.

Answer $\begin{bmatrix} -4 & 13 \\ -6 & -16 \end{bmatrix}$

Question 5

Check whether the matrix $A = \begin{pmatrix} 4 & 8 & 6 \\ 11 & -6 & 5 \\ 9 & -4 & 7 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 6 & 8 \\ 4 & 7 & 9 \end{pmatrix}$ can be multiplied? Justify your answer.

Solution

order of $A = 3 \times 3$

Order of $B = 2 \times 3$

Inner dimensions do not match so this multiplication is not possible.

Question 6

Find the inverse of $A = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$

Answer

$$\begin{bmatrix} -4/2 & 2/2 \\ -3/2 & 1/2 \end{bmatrix}$$

Question 7

Find the determinant of the following matrix

$$A = \begin{bmatrix} -3 & 4 & 9 \\ 6 & -1 & -5 \\ 2 & 8 & 7 \end{bmatrix}$$

Answer: 143

Question 8

Given the sequence, 150, 300, 450, 600, ..., find the 15th term.

Solution

$$a = 150, d = 150, n = 15$$

$$a_n = a + (n-1)d$$

$$a_{15} = 150 + 150(14) = 2250$$

Question 9

What is the general term of the sequence 1, 8, 27, 64, 125

Solution

$$a_n = n^3$$

Question 10

The 3rd and the 4th terms of a geometric progression are 18 and 6 respectively. Find the sum to infinity of the progression.

Solution

$$\text{Given that } ar^2 = 18 \text{ and } ar^3 = 6$$

$$r = 6/18 = 1/3.$$

$$a(1/9) = 18$$

$$a = 162$$

$$S_{\infty} = \frac{a}{1-r} = 243$$

Question 11

If first term $a = 4$, common difference $d = 2$, then what is the value of 10th term in an arithmetic sequence?

Solution

$$\begin{aligned} a_n &= a + (n-1)d \\ a_{10} &= a + 9d \\ &= 4 + 9(2) \\ &= 22 \end{aligned}$$

Question 12

Suppose $x + 2a, x + 4a, x + 6a, \dots$, is an arithmetic sequence. Write its corresponding arithmetic series.

Hint:

Just sum (add) all the terms to find the corresponding series.

Question 13

If $n = 1, 2, 3, 4, 5, 6, \dots$ then find the first four terms of the sequence $a_n = (n + 2)^2$.

Solution

$$\because n = 1, 2, 3, 4, 5, 6, \dots$$

$$\because a_n = (n + 2)^2$$

$$\left. \begin{aligned} \text{For } n = 1 &\Rightarrow a_1 = (1 + 2)^2 = (3)^2 = 9 \\ \text{For } n = 2 &\Rightarrow a_2 = (2 + 2)^2 = (4)^2 = 16 \\ \text{For } n = 3 &\Rightarrow a_3 = (3 + 2)^2 = (5)^2 = 25 \\ \text{For } n = 4 &\Rightarrow a_4 = (4 + 2)^2 = (6)^2 = 36 \end{aligned} \right\}$$

Question 14

Determine whether the sequence 9, 11, 13, 15, 17, ... is arithmetic or geometric. Give reason for your answer.

Solution

First find the difference between each pair of consecutive terms.

$$11 - 9 = 2$$

$$13-11 = 2$$

$$15-13 = 2$$

$$17-15 = 2$$

$$\text{Common difference} = d = 2$$

The difference between the consecutive terms of the sequence is constant. The constant difference is called common difference. Hence the sequence is arithmetic.

Question 15

Find the 5th term of the geometric sequence with $a_1 = 20$ and $a_2 = 40$.

Solution

$$r = \frac{2^{\text{nd}} \text{ term}}{1^{\text{st}} \text{ term}} = 40 / 20 = 2$$

$$a_n = ar^{n-1}$$

$$a_5 = 20 \cdot (2)^4 = 20 \cdot 16 = 320$$

Question 16

If $a_0 = 3$, $a_1 = 4$, $a_2 = -2$, and $a_3 = 1$, then compute the summation: $\sum_{i=0}^2 2a_i$

Solution

$$\because a_0 = 3, a_1 = 4, a_2 = -2, \text{ and } a_3 = 1$$

$$\begin{aligned} \Rightarrow \sum_{i=0}^2 2a_i &= 2(a_0 + a_1 + a_2) \\ &= 2(3 + 4 - 2) \\ &= 2(5) \\ &= 10 \end{aligned}$$

Question 17

What is the sigma notation for the series: $2(1)^2 + 2(2)^2 + 2(3)^2 + \dots + 2(10)^2$?

Question 18

If $a_0 = 3$, $a_1 = 2$, $a_2 = 5$, $a_3 = 4$ and $a_4 = -9$, then compute the summation: $\sum_{i=0}^3 3a_i$?

Question 19

Compute the summation: $\sum_{n=0}^3 (3n^2 + 1)$.

Question 20

Formula for the sum of infinite geometric series is Whose first term is 'a' and common ratio is 'r'.

Question 21

Find the sum of first 'n-1' natural numbers.

Practice Questions for Lecture 16 to 22

Question 1

A teacher is making a multiple-choice quiz. She wants to give each student the same questions, but have each student's questions appear in a different order. If there are 32 students in the class, what is the least number of questions the quiz must contain?

Solution

If there were two questions on the quiz, we could prepare two quizzes with the questions in different order, $2 \cdot 1 = 2$.

If there were three questions, we could get $3 \cdot 2 \cdot 1 = 6$ different orders.

If there were four questions, we could get $4 \cdot 3 \cdot 2 \cdot 1 = 24$ different orders -- not quite enough for the class of 27 students.

If there were five questions, we could get $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ different orders. The teacher will need at least 5 questions on the quiz.

Question 2

The local Family Restaurant has a daily breakfast special in which the customer may choose one item from each of the following groups:

Breakfast Sandwich	Accompaniments	Juice
--------------------	----------------	-------

egg and chicken egg and beef egg and cheese egg and mutton	breakfast potatoes apple slices fresh fruit cup pastry plain cake	orange cranberry tomato apple grape
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How many different breakfast specials are possible?

Solution

Basic counting principle: Sandwiches x Accompaniments x Juice
 $= 4 \cdot 5 \cdot 5 = 100$ breakfast choices.

Question 3

The local Family Restaurant has a daily breakfast special in which the customer may choose one item from each of the following groups:

Breakfast Sandwich	Accompaniments	Juice
egg and chicken egg and beef egg and cheese egg and mutton	breakfast potatoes apple slices fresh fruit cup pastry plain cake	orange cranberry tomato apple grape

How many different breakfast specials without meat are possible?

Solution

Meatless means that under Sandwiches there will be only one choice.
 Sandwiches x Accompaniments x Juice: $1 \cdot 5 \cdot 5 = 25$ meatless breakfast choices.

Question 4

If in first bag there is a red and blue and orange ball and in second bag there is green, black, and yellow ball. You draw one ball from first bag and then one ball from second bag. How many different pairs of ball can you have?

Solution

According to multiplication rule there are $3 \cdot 3 = 9$ pairs of ball can be drawn.

Question 5

Suppose you have 5 pictured cards that have the pictures of the letters A, B, C, D and you want to arrange them in a row to form “words”. How many 4-letter words are there?

Solution

Here we are arranging five distinct letters in a line.

The number of permutations is $4! = 24$

Question 6

How many ways are there to sit 6 people around a circular table, where seatings are considered to be the same if they can be obtained from each other by rotating the table?

Solution

First, place the first person in the north-most chair. This has only one possibility. Then place the other 5 people.

There are $5P5 = 5! = 120$ ways

By the product rule, we get $1 \times 120 = 120$.

Alternatively,

There are $6P6 = 720$ ways to seat the 6 people around the table.

For each seating, there are 6 “rotations” of the seating.

Thus, the final answer is $720/6 = 120$.

Question 7

How many 2-permutations are there of $\{A, B, C, D\}$? Write them all.

Solution

Number of 2-permutation of 4 elements is

$$\therefore {}^nP_r = P(n, r) = \frac{n!}{(n-r)!}$$

$$\begin{aligned}\Rightarrow {}^4P_2 &= P(4, 2) = \frac{4!}{(4-2)!} \\ &= \frac{4!}{2!} \\ &= 12\end{aligned}$$

These 12 permutations are:

AB, AC, AD, BA, BC, BD, CA, CB, CD, DA, DB, DC

Question 8

Find the number of distinct permutations of the letters of the word 'CHEER'.

Solution

The total number of letters = 5,

Number of C's = 1, Number of H's = 1, Number of R's = 1, Number of E's = 2.

$$\begin{aligned}\text{The total number of distinct permutations} &= \frac{5!}{1!1!1!2!} \\ &= \frac{120}{2} \\ &= 60\end{aligned}$$

Question 9

There are three juniors and three seniors in the Service Club. The club is to send four representatives to the State Conference. How many different ways are there to select a group of four students to attend the conference?

Solution

Choose 4 students from the total number of students. Order is not important. There are 6 total students. So number of choices is ${}^6C_4 = 15$

Question 10

There are four juniors and three seniors in the Service Club. The club is to send three representatives to the State Conference. If the members of the club decide to send one junior and two seniors, how many different groupings are possible?

Solution

Choose 1 junior and 2 seniors: $4C_1 \cdot 3C_2 = 12$

Question 11

Suppose that $X = \{a, b, c\}$. Then list all 3-combinations of the set X , and hence find the value of $C(3,3)$.

Solution

$X = \{a, b, c\}$.

Then 3-combinations of the 3 elements of the set X is: $\{a, b, c\}$.

Hence $C(3,3) = 1$.

Question 12

If $W = \{x, y, z, t\}$. Then list all 3-combinations of the set W , and hence find the value of $C(4,3)$.

Solution

$W = \{x, y, z, t\}$.

Then 3-combinations of the 4 elements of the set X are:

$\{x, y, z\}$, $\{x, y, t\}$, $\{x, z, t\}$ and $\{y, z, t\}$.

Hence $C(4,3) = 4$.

Question 13

Compute: $C(9, 6)$.

Solution

$$\therefore {}^nC_r = \frac{n!}{(n-r)! r!}$$

$$\Rightarrow {}^9C_6 = \frac{9!}{(9-6)! 6!}$$

$$= \frac{9!}{3! 6!}$$

$$= \frac{362880}{6 \cdot (720)}$$

$$= \frac{362880}{4320}$$

$$= 84$$

Question 14

There are total 14 players in a cricket team. The captain needs to choose 10 players to play in a match. How many possible teams can be chosen?

Solution

This is a combination problem as the order in which the teams are chosen is not important. The number of combinations is

$$\therefore {}^nC_r = \frac{n!}{(n-r)! r!}$$

$$\Rightarrow {}^{14}C_{10} = \frac{14!}{(14-10)! 10!}$$

$$= \frac{14!}{4! 10!}$$

$$= \frac{87178291200}{24 \cdot 2628800}$$

$$= 1001$$

Question 15

In how many ways can a team of 5 candidate be formed out of total 12 candidates such that two particular candidates should always be included in team formed?

Solution

Two particular candidates should always be included in team.

i.e., we have to select $5-2 = 3$ persons from $12-2 = 10$ persons

Hence, the required number of ways are:

$$\begin{aligned}\therefore {}^nC_r &= \frac{n!}{(n-r)! r!} \\ \Rightarrow {}^{10}C_3 &= \frac{10!}{(10-3)! 3!} \\ &= \frac{10!}{7! 3!} = 120\end{aligned}$$

Question 16

Find the first two terms in the expansion of $(4 - x)^6$ in ascending powers of x .

Solution

$$\begin{aligned}(4-x)^6 &= \binom{6}{0}(4)^6(-x)^0 + \binom{6}{1}(4)^5(-x)^1 + \binom{6}{2}(4)^4(-x)^2 + \binom{6}{3}(4)^3(-x)^3 + \binom{6}{4}(4)^2(-x)^4 + \binom{6}{5}(4)^1(-x)^5 + \binom{6}{6}(4)^0(-x)^6 \\ &= 4096 - 6144x + 3840x^2 + \dots\end{aligned}$$

Check first two terms only which are $4096 - 6144x$

Question 17

Find the value of the constant a for which the coefficient of x in the expansion of $(1 + ax)(2 - x)^6$ is 128.

Solution

$$(1 + ax)(2 - x)^6 = (1 + ax)(64 - 192x + 240x^2).$$

Collecting the terms only, we get,

$$-192x + 64ax = (64a - 192)x.$$

Since the coefficient of x is given to be 64, we get

$$64a - 192 = 128a$$

$$a = 5$$

Question 18

Calculate the coefficient of x^8y^3 and xy^{10} in the expansion of $(x+y)^{11}$

Solution

$$11C8 = 165$$

$$11C1 = 11$$

Question 19

Write the distance formula to find distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$.

Solution

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Question 20

What happens if the slopes of two lines are equal?

Solution

The two lines are parallel.

Question 21

Find the midpoint of a line whose coordinates are A (4, 8) and B (8, 6).

Solution

The midpoint of AB has coordinates

$$\left(\frac{4+8}{2}, \frac{8+6}{2} \right) = (6, 7)$$

Question 22

Find the Slope of the line through the points (2, 2) and (6, 4)

Solution

$$Slope = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4-2}{6-2} = \frac{2}{4} = \frac{1}{2}$$

